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Functionally graded TPU gyroid structures for cushioning applications

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ABSTRACT

Three dimensional (3D) printed, thermoplastic polyurethane (TPU) gyroid structures are proposed as an alternative to polyurethane medical foams used in wheelchair cushions. Compared to traditional medical foams, 3D printed foam cushions can be made with highly customized shape and stiffness, which is hypothesized to enable improvements in pressure offloading and pressure injury prevention. However, to realize these benefits, the mechanical behavior of 3D printed foams must be determined. To this end, this work investigates the mechanical behavior of TPU gyroid foams with variable stiffness and explores how stiffness gradients can be leveraged to produce 3D printed foams with superior behavior to their polyurethane counterparts. We show that the stiffness gradient of 3D printed gyroid foams can be effectively manipulated to control cushion deformation and prevent “bottoming out”. Graded gyroid structures demonstrated improved compression fatigue resistance over uniform gyroid structures for stiffness gradients in both the direction of load and perpendicular to the direction of load. Fatigue resistance was improved in gyroids with moderate gradients and smooth stiffness transitions over sudden changes. These findings will inform the design of wheelchair cushions with functionally graded metamaterials, further improving understanding of how they may be applied in design for pressure injury prevention.

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1. Introduction

Pressure injuries occur due to intense or prolonged pressure on soft tissue resulting in localized soft tissue [1,2]. Wheelchair users are particularly vulnerable to pressure injuries [1,3–7]. In addition to their significant impact on quality of life, pressure ulcers are the most common cause for inpatient hospitalization of people with spinal cord injuries [8]. Thus, improved prevention of pressure ulcers will provide tangible benefits in terms of quality of life and reduce hospitalization for wheelchair users.

One of the best ways of reducing pressure injuries is the use of cushions that reduce the peak surface pressures typically occurring under bony prominences [1,3,9]. Traditionally, the most effective wheelchair cushions are custom contoured to the user and are manufactured from polyurethane foams that envelop the buttocks to maximize surface area and reduce pressure [10,11]. Clinicians often use multilayered polyurethane foams, with softer foams on top to redistribute pressure while a firmer foam underneath provides durability and support [12]. A cushion is ‘bottomed-out’ when an increase in load does not produce an increase in deflection [13], which is an issue more likely to be faced by users with high body weights when using off-the shelf cushions [14]. Therefore, a well design offloading cushion should, effectively redistribute

pressure, not-bottom out, and be durable and long lasting. Major limitations with these existing foam cushions include expense, material waste and extensive manufacturing time, as each cushion is custom manufactured for the patient, and may require multiple consults [10,15].

3D printed metamaterial wheelchair cushions offer the potential to address these challenges, providing better offloading through tuned mechanics, and improved product-life through open-cell design which enables airflow and ease-of-cleaning. 3D printed cushions could be custom made based on a 3D scan of the patient’s anatomy [16], potentially reducing the number of clinic visits and providing better fitting. This could provide significant improvements to user quality-of-life and reduced burden on the healthcare system. Functionally graded metamaterials (FGMs) are a class of lattice structures consisting of repeating cells whose properties, like stiffness, can be controlled by manipulating geometric properties like lattice wall thickness or cell length [17–19]. Additionally, FGMs tend to exhibit higher potential for energy absorption [17,20,21]. In medical offloading applications FGMs offer the potential for increased customizability compared to traditional methods. Since 2018, several studies have investigated using FGMs to design patient-specific insoles for offloading and prevention of diabetic foot ulcers with promising results [22–25].

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The popularity of using flexible metamaterials as an alternative for foam has grown in recent years. TPU gyroid structures have been shown to have a similar stress-strain response to typical pressure relieving foams [26,27]. This result was confirmed by Holmes et al. in 2022 who established that TPU gyroid structures exhibit similar compressive behavior and low cycle fatigue characteristics to rehabilitation foams typically used in wheelchair cushions [28]. The popularity of gyroids for this application is attributed to the significant volume of research characterizing them for their enhanced energy absorption [29], printability [30] and fatigue characteristics [31]. As of early 2023, some additive manufacturing companies have announced they can now produce 3D-printed TPU wheelchair cushions which are temperature-regulating and fast-drying [32], as well as contoured cushions which can combine soft, moderate, and firm stiffnesses using functional grading [33]. However, there remains a noticeable gap in the literature regarding the impact of functional stiffness grading on compressive behavior in the context of wheelchair cushions. The application and effectiveness of 3D printed foams in cushion design is severely limited until this gap is addressed.

Previous investigations on FGMs or flexible metamaterials have focused on uniaxial compression tests [17,20,26,28,29,34]. Uniaxial compression testing of non-flexible metamaterials reveals non-graded structures will deform more uniformly than structures with a stiffness grading in the direction of applied load, where lower stiffness regions will deform before higher stiffness regions [17,20]. Stiffness grading lateral to the direction of load has also been used to control how load is carried across a structure in graded metamaterials [20], which is hypothesized to enable control over pressure distribution seen across a wheelchair cushion. Stiffness grading has also been shown to improve fatigue resistance in rigid metamaterials when tested at low strains (up to 0.25%) [31]. However, the effect of stiffness grading on fatigue resistance in flexible metamaterials when tested to high strains is less well studied. TPU exhibits softening behavior under cyclic loading where larger strains produce greater softening and stress-strain behavior typically stabilizes after four loading cycles [35]. Under high cycles, hysteresis induced self-heating may further amplify softening [28,35]. Thinner walled structures also have lower fatigue resistance than thicker walled structures [36], and because flexible graded metamaterial structures are expected to deform less evenly than non-graded structures, this may impact fatigue life through impacting the strain-softening response in the TPU. For example, when considering a graded TPU metamaterial under high compressive strain, a thin-walled region would collapse first, causing high strain in the parent material in that region and potentially impacting fatigue life compared with a uniform TPU metamaterial.

Compression testing of lattice structures reveals the outer walls contribute less to the overall strength than the inside walls, leading to edge effects [37]. Size effect analysis for cellular materials determines the minimum number of unit cells required in a sample before the impact of edge effects is negligible to the overall result [37–39]. The minimum number of

cells required is impacted by the base material and cell structure [37–39]. Previous studies testing TPU gyroid structures have been on cubic or cylindrical samples of 1–5 cells [19,27–29,40]. The scalability of these type of results, however, is not well understood. Without considering edge effects, the softness of 3D printed foams might be overestimated, limiting how effectively they can be translated to cushion design. Additionally, the loading applied to gyroid structures in the context of a wheelchair cushion cannot be fully represented by only uniaxial compression testing. As such, further material characterization of flexible FGMs in the context of their performance in wheelchair cushions is needed.

This article shows that functionally graded, flexible lattice structures are mechanically suitable for offloading applications through enabling control over pressure distribution across a cushion. While non-graded flexible metamaterials have been previously investigated [28], this work is the first demonstrating experimental testing of functionally graded, flexible metamaterials as an alternative to rehabilitation foam in wheelchair cushions for pressure injury prevention with novel offloading capabilities. This involved the design and manufacture of functionally graded gyroids to be tested in compression to establish their stress–strain, energy absorption and fatigue resistance. Section 2 describes sample selection, design and manufacture as well as the testing methodologies used in this study. The specific novelty of this work centers on cyclic loading of structures with stiffness grading both in the direction of load and perpendicular to the direction of load. This type of testing is required to understand the mechanical properties of flexible gyroid structures with stiffness grading in all three dimensions as could ideally be applied in a 3D printed wheelchair cushion. This work also measures the scalability of material testing performed on smaller scale TPU gyroid to test if these are translational into cushion design. The results from each step are presented and discussed in Section 3, while Section 4 summarizes the conclusions and implications for future research and applications.

2. Materials and methods

2.1. Mathematical description of graded and uniform gyroids

The lattice structure investigated in this work is a sheet gyroid structure, stemming from the following mathematical definition of the triply periodic minimal surface (TPMS):

$$\sin(x)\cos(y) + \sin(y)\cos(z) + \sin(z)\cos(x) = 0 \quad (1)$$

where x , y , and z are Cartesian coordinates. Where Eq. (1) describes a zero-thickness surface, the sheet gyroid structure may also be expressed as Eq. (2) where l and t represent the unit-cell length and thickness respectively.

$$\begin{aligned} \sin\left(\frac{2\pi}{l}x\right)\cos\left(\frac{2\pi}{l}y\right) + \sin\left(\frac{2\pi}{l}y\right)\cos\left(\frac{2\pi}{l}z\right) \\ + \sin\left(\frac{2\pi}{l}z\right)\cos\left(\frac{2\pi}{l}x\right) = t \end{aligned} \quad (2)$$

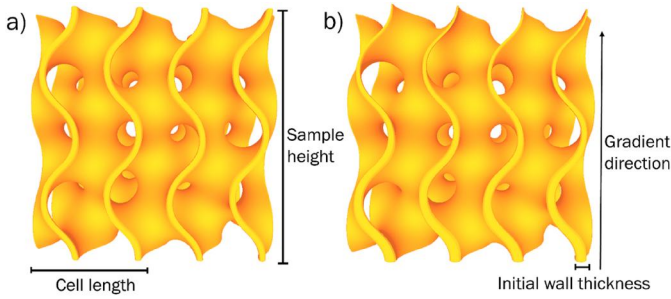


Figure 1. Labeled comparison of a) uniform and b) functionally graded gyroids (varying wall thickness).

By replacing the constant l or t variables as functions in terms of x , y , and/or z , a structural gradient is formed [21]. In this work, thickness is varied in the z -direction *via* the following linear relation:

$$t(z) = \frac{t_{final} - t_{initial}}{distance} z + t_{initial} \quad (3)$$

where $t_{initial}$ is the wall thickness at the base of the structure and t_{final} is the wall thickness at the top of the structure as shown in Figure 1. This gradient ranges the whole lattice structure, meaning *distance* is the sample height (range of z). By changing the initial and final thicknesses across the structure, the “steepness” of the gradient could be varied between samples. This metric was used to generate samples for different tests.

The solid volume fraction (SVF) of a metamaterial describes its relative density and can be described as follows where V_{wall} and V_{void} are the volumes of the wall and void, respectively. SVF increases approximately linearly with wall thickness when cell size remains constant.

$$SVF = \frac{V_{wall}}{V_{wall} + V_{void}} \quad (4)$$

To design the geometries for manufacture, stereolithography (STL) files of each design were produced with a custom C# .NET software tool. The STLs were processed in MeshLab (v2021.05, MeshLab) to determine V_{wall} and V_{void} , respectively [41].

2.2. 3D printed sample fabrication

Samples were fabricated *via* fuze filament fabrication (FFF) following a previously established methodology [28]. In this, STL models were converted to toolpaths using PrusaSlicer (v2.4.1, Prusa Research a.s., Czechia), and fabricated on a Prusa i3 MK3S printer (Prusa Research a.s., Prague, Chechia) which had been retrofitted with a flexion extruder coupled with an E3D v6 HotEnd (E3D-Online Ltd, England). Samples were fabricated with NinjaFlex filament (NinjaTek, USA), which has a density of 1.19 g/cm^3 density and 85 A shore hardness [42]. The fabricated SVF was determined based on the mass of each sample, and the filament density.

2.3. Investigating the impact of gradient steepness

To understand the impact of gradient steepness on FGM deformation and load distribution characteristics, uniaxial compression was conducted on the FGM samples. Samples were compressed along the gradient axis, with the thicker ($t_{initial}$) wall thickness at the base and thinner (t_{final}) wall thickness on the top contacting the compression plate. Five different 70 mm cubic samples were tested in this manner, with a mean SVF of 15%, but varying the steepness of the thickness gradient across the structure and one non-graded sample was included as a control. There were three replicates for each sample. A unit cell size of 35 mm was used for all with the full specimen details given in Table 1 and Figure 2. Compression testing was conducted using an Instron 5967 Universal Testing Machine (Illinois Tool Works Inc., USA) with a 30 kN load cell. All samples were fabricated successfully, free of visual defects, as demonstrated in Figure 2. A small deviation was recorded between the targeted and fabricated SVF as detailed in Table 1.

2.3.1. Stress-strain characterization

Stress-strain response was measured in accordance with the international standards for Polymeric materials, cellular flexible – Determination of stress-strain characteristic in compression – Part 1: Low-density materials BS EN ISO 3386-1:1997 + A1:2010 [43]. This guided the selection of sample size, which were manufactured in 70 mm cubes to meet the minimum side length of 50 mm and also capture two unit cells of 35 mm length. Samples were stored according to this standard, remaining uncompressed for at least 16 h before testing in a humidity-controlled cabinet set at 50% relative humidity. For each different geometry to be tested, three samples were manufactured with the average of these results used for analysis. A compression rate of 100 mm/min was applied until a strain of 70% was reached. Each sample was loaded and unloaded with four repeats and the final cycle was recorded for analysis. A preload of 5 N was used to maintain constant contact between the sample and compression plate between cycles.

2.4. Design of a functionally graded gyroid to replicate multilayered rehabilitation foams

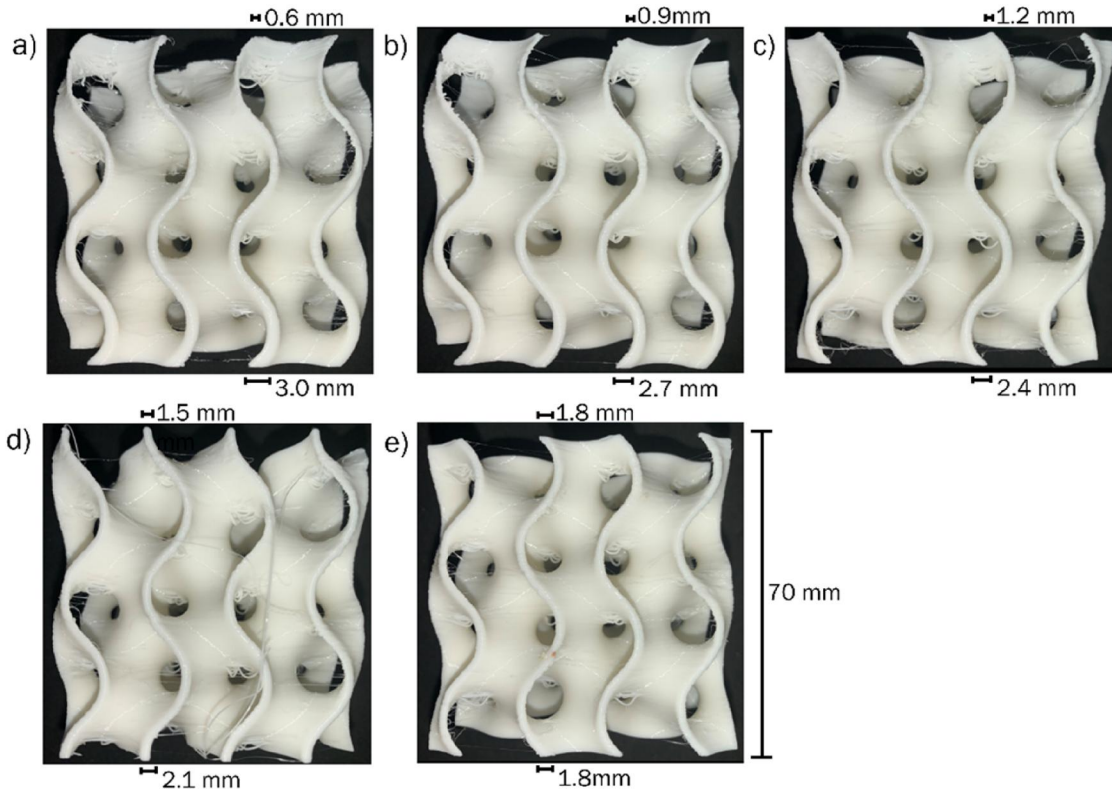
In this section we explore the compressive properties of graded metamaterials, compared to layered foam structures (e.g. Figure 4c). Our previous work determined target SVF for non-graded NinjaFlex TPU gyroid structures with comparable properties to common rehabilitation foams (shown in Table 2) [28]. The following soft rehab foams manufactured by Dunlop Foams, Australia, reflect soft, medium, and hard foams, respectively,¹ and were used as a benchmark for this article: Green EN36-90, Yellow EN40-230, and Pink MA35-600 foam [12]. Based on this, wall thicknesses were determined to achieve these target SVFs and are presented

¹Classification of soft, medium and hard taken from rehabilitation engineer standard classification.

Table 1. Summary of sample design parameters and variations between designed and fabricated samples for gradient steepness study.

Sample	Initial thickness (mm) (targeted)	Final thickness (mm) (targeted)	Initial SVF (targeted)	Final SVF (targeted)	Mean SVF (targeted)	Mean SVF (fabricated)
5-25%	3	0.6	25.02%	5.05%	15.04%	14.40%
8-21%	2.7	0.9	22.62%	7.66%	15.14%	14.42%
10-20%	2.4	1.2	20.19%	10.21%	15.20%	14.64%
13-18%	2.1	1.5	17.73%	12.73%	15.23%	14.41%
15% (uniform)	1.8	1.8	15.24%	15.24%	15.24%	14.41%

With “initial” referring to the bottom of the sample, and “final” referring to the top.

**Figure 2.** Test samples of functionally graded gyroids with SVF varying between a) 5–25%, b) 8–21%, c) 10–20%, d) 13–18% and e) constant 15% SVF.**Table 2.** Theoretical design requirements for TPU gyroids to mechanically replicate traditional rehabilitation foams.

Equivalent rehab foam	Target SVF NinjaFlex Gyroid [28]	Target wall thickness with 30 mm cell length (mm)
Green EN36-90	4.81%	0.48
Yellow EN40-230	6.91%	0.69
Pink MA35-600	8.42%	0.84

in Table 2. For this section, a cell length of 30 mm was used, such that the total sample height could be more easily divided into three for assembling three-layer foam structures.

A multilayered comparison rehabilitation foam sample of a 60 mm cube was designed using even sections of three standard rehabilitation foams (Green EN36-90, Yellow EN40-230 and Pink MA35-600) inspired by the design of custom wheelchair cushions from firmest at the base to softer at the top. Next, two functionally graded gyroids were designed with the goal of replicating the compressive behavior of the multilayered foam. The first gyroid consisted of three distinct stiffness sections defined by the three wall thickness selections to match the three distinct sections of

foam in the multilayered foam samples as shown in Table 2. The second gyroid design had a smooth linear transition in wall thickness of the total height, with the wall thickness associated with each foam aligning with the midpoint of that stiffness region. For example, when considering the wall thickness gradient over the height of this gyroid, the wall thickness was 0.84 mm aligned with the mid-point of the Pink MA35-600 region of the foam sample and became firmer below that. As 0.48 mm was the thinnest printable wall thickness with the selected print settings, the gradient could not be continued all the way through to the top of the structure. A summary of each of these samples is given in Figure 3. Three repeats of each sample were manufactured, and stress-strain characteristics were derived using the same testing as the previous compression tests in this work.

2.5. Investigating the scalability of material testing results

Many previous studies of TPU gyroid structures adhere to the international standards for Polymeric materials, cellular flexible—Determination of stress-strain characteristic in

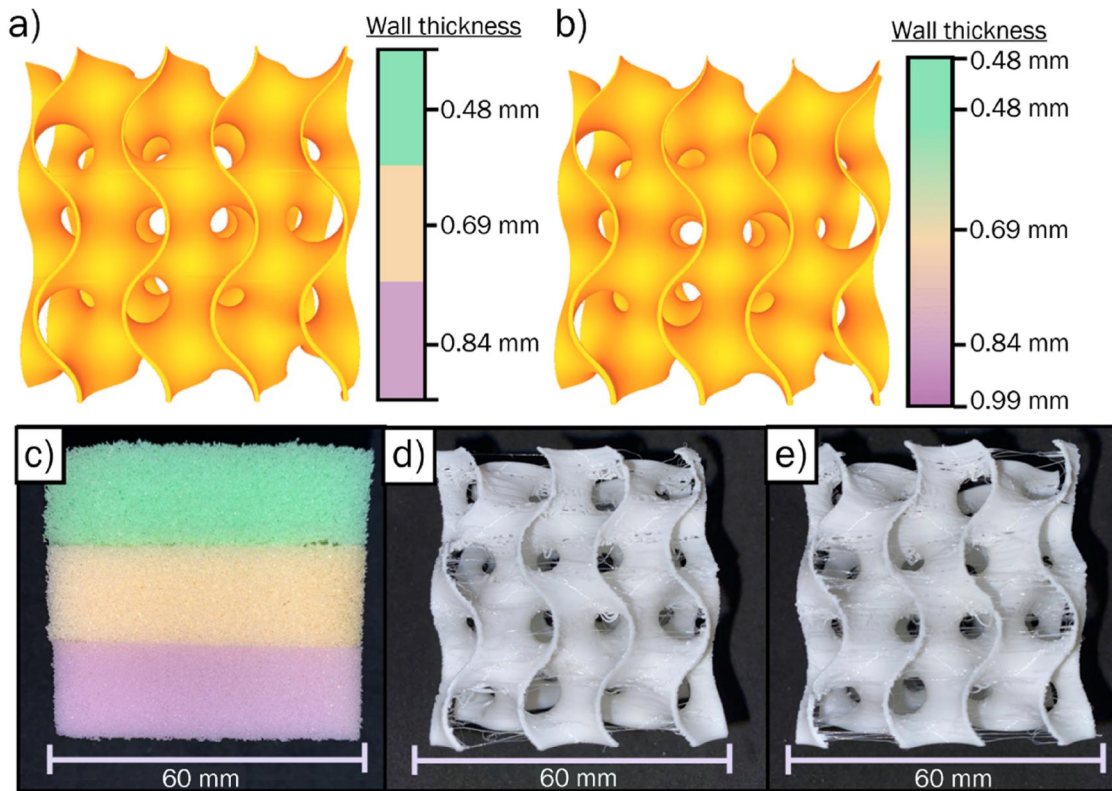


Figure 3. a) Gyroid design with three separate stiffness regions, b) gyroid design with linear wall thickness gradient, c) multilayered foam sample, d) printed gyroid with three separate stiffness regions and e) printed gyroid with linear wall thickness gradient to replicate multilayered foam.

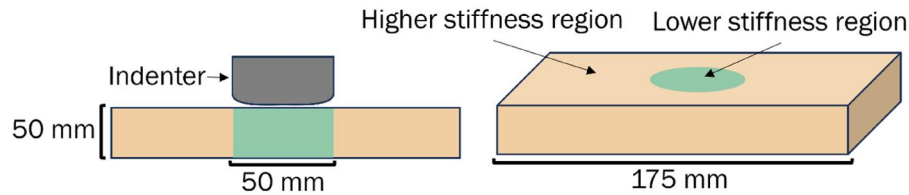


Figure 4. a) Dimensioned diagram of the indentation testing setup as viewed from the mid-plane (left) and angle (right). b) example of the experimental setup for indentation testing when at maximum compression.

compression—Part 1: Low-density materials BS EN ISO 3386-1:1997 + A1:2010 [19,28,29,43]. The minimum sample length required by the standard is 50 mm. For cushioning applications such as in this work, this may mean that only two unit cells are represented. It is hypothesized that this standard is appropriate for identifying trends in the behavior of mechanical metamaterials, but that the stress-strain curve is not necessarily scalable due to edge effects. A test was constructed to determine the minimum number of unit cells required to obtain a representative result.

Compression testing was conducted using an Instron 5967 Universal Testing Machine (Illinois Tool Works Inc., USA) with a 30 kN load cell. Outside of specimen dimensions, all other aspects of the standard were followed, including testing in triplicate. Sample design for scalability testing was constrained by equipment limitations, where the maximum edge length for a cuboid sample was 90 mm. The unit cell length was decreased to 10 mm so that more unit cells could be captured within this size constraint. A wall thickness of 0.6 mm and SVF of 17.84% was chosen.

First, samples were designed to measure the impact of the height based on the number of cells in the direction of load. This has relevance in contoured cushion design where the depths of the contour will have less unit cells underneath the user than the other regions of the cushion. These samples were 7×7 unit cells wide and 1, 2, and 3 unit cells high. The next samples were designed to determine the minimum number of unit cells wide a sample must be for the result to be scalable. All samples were 2 unit cells high and the widths were 2×2 , 3×3 , 5×5 , 7×7 , and 9×9 .

2.5.1. Uniform compression fatigue

Previous studies on FGMs show gradient steepness impacts deformation behavior, however the implications of this for the compressive fatigue resistance of such materials is unknown and may impose constraints on the design of off-loading medical devices utilizing flexible FGMs [30–32,44]. The low cycle fatigue response of some of the previous cubic samples with varying steepness gradients was tested using an Instron Electropuls E10000 Linear-Torsion Fatigue Testing

Machine with a 3kN load cell (Illinois Tool Works Inc., USA). The orientation for testing was kept the same as previous stress-strain testing in this study with the higher thickness at the base of the subject. Samples representing steep and moderate gradients were tested for 10,000 cycles each and compressed from 10 to 40% strain in a triangular waveform at a frequency of 4 Hz. At the end of the 10,000 cycles, an infrared image was taken of the samples using a FLIR One Pro camera (Teledyne FLIR LLC., USA) to measure any temperature change. The samples were left for 24 h to return to room temperature then tested again.

2.5.2. Indentation fatigue

Indentation testing of foam cushions is used to predict loss of hardness and thickness caused by continuous use and more accurately represents the loading on a cushion compared to uniaxial compression. The procedure from the international standard, “Flexible cellular polymeric materials – Determination of fatigue by constant-load pounding BS EN ISO 3385:2014”, was modified to accommodate for the limitations of the fatigue testing machine available. A test sample of 175×175 mm length and 50 mm height was compressed by a cylindrical indenter of 50 mm diameter with edges. A frequency of 4 Hz was used and strokes oscillated in a triangular waveform between 10 and 40% depth. Figure 4 shows the typical setup for the cyclic indentation testing used in this study.

After 10,000 cycles, an infrared image was taken of the samples using a FLIR One Pro camera (Teledyne FLIR LLC., USA) to measure any temperature change. The samples were left for 24 h to return to room temperature then tested again.

Motivated by wheelchair cushions which offload strain from-high risk areas by being softer under boney prominences

and firmer around the outer buttocks, three cases were considered which explored how lateral stiffness transition impacts fatigue durability. The first case was representative of a cushion with a soft region under a bony prominence and firmer around the outer buttocks to offload strain from high-risk areas, with an instantaneous stiffness transition. The second was like the first, but with a graded transition of wall thickness. The third case considered a cushion of uniform stiffness. These cases are shown in Figure 5a–c.

Using a cell size of 35 mm, wall thickness in the soft region was 0.48 and 0.81 mm in the stiff region. These respective wall thicknesses correspond with SVFs which are comparable to existing “soft” and “firm” rehabilitation foams at 4.81% (EN36-90) and 6.91% (MA35-600). This allowed for a similar range in stiffnesses as is currently used in wheelchair cushions manufactured from rehabilitation foam. As a control, a uniform cushion was designed with a constant wall thickness of 0.81 mm and a foam, EN36-90 sample was also cut to the same dimensions. Two gradient types were designed to test: one with a linear transition from 0.48 at the center of the circle increasing to a maximum of 0.81 mm at the outside radius achieved by replacing wall thickness in the gyroid Eq. (2), with a function representing thickness in the form of the equation for the height of a cone (Eq. 5).

$$t(x, y) = \frac{t_{final} - t_{initial}}{outer\ radius} * \sqrt{x^2 + y^2} + t_{initial} \quad (5)$$

Here, $t_{initial}$ is the wall thickness at the center point of the cylinder, t_{final} is the wall thickness at the outer radius. The other indentation sample was designed with a soft region in the center of the sample aligning with where it would be in contact with the indenter and was firmer surrounding this area. This softer region had a wall thickness of 0.48 mm in

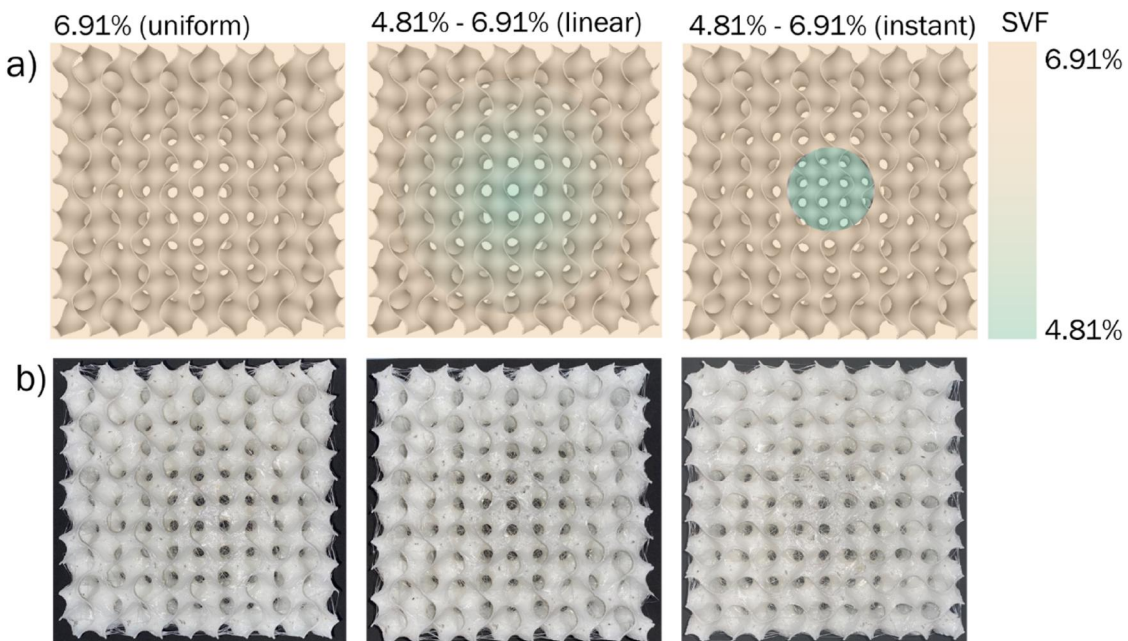


Figure 5. The top row shows CAD models of each sample with wall thickness change color coded across the structure while the bottom row shows the corresponding print result where all sample lengths are 175 mm long. a) uniform 0.81 mm wall thickness (6.91% SVF), b) linear increase in all thickness from 0.48 - 0.81 mm (4.81–6.91% SVF) and c) instant change in wall thickness from 0.48 to 0.81 mm (4.81 to 6.91% SVF).

the center which then instantly increased to 0.81 mm surrounding that region. The variation in wall thickness across the samples as designed is shown in Figure 5, along with the printed samples.

3. Results and discussion

3.1. Fabrication

Samples were fabricated successfully with no significant observed defects, though some minor defects such as stringing and gaps were observed where the gyroid structure had regions of bridging. Defects were more prominent in the regions with lower wall thickness. Further detail is provided in Appendix A.

3.2. Impact of gradient steepness

For manufactured lattices of similar total weight and overall SVF, the stiffness gradient magnitude impacts the compressive deformation behavior. This is demonstrated in Figure 6a, where the sample with a steep SVF gradient deforms by distinct sequential layer collapse from the top down. For the more moderate gradient, Figure 6b, the deformation is a combination of sequential layer collapse initially, followed by a more uniform deformation. The uniform lattice in contrast deforms similarly across the total height of the structure. The stress-strain response of each differing gradient

sample Figure 6c also shows this behavior wherein structures of larger gradients do not demonstrate plateau stress or top-down sequential layer collapse like those of lower gradient to uniform density. Vertical stiffness gradients have traditionally been used in the design of wheelchair cushions by layering different stiffness foams to achieve a desired compressive response. The results here demonstrate that vertical stiffness grading can be used to manipulate the behavior of flexible gyroids to a wide range of different behaviors which would translate to cushioning. A higher stiffness gradient will result in lower stresses at low strains but higher stresses at higher strains as compared to uniform structure. In the context of a cushion this may be likened to it being softer during the act of sitting down but firm enough, so the cushion does not “bottom out.” Alternatively, uniform and low gradient structures will tend to continuously compress under constant load until densification occurs under the user. The impact of steepness gradients on stress-strain response also influences how each sample absorbs energy when compressed. Figure 6b shows although all samples had similar energy absorption over the full compression to 75% strain, at lower strains up until 25% compression, energy absorption increased as stiffness gradient decreased. At higher strains, such as between 50 and 75% compression, the samples with a higher stiffness gradient tended to absorb more energy. This supports what can be observed in Figure 6c where the compressive stress-strain response can be shifted through vertical stiffness gradient. In practice, this means a softer initial compression when sitting

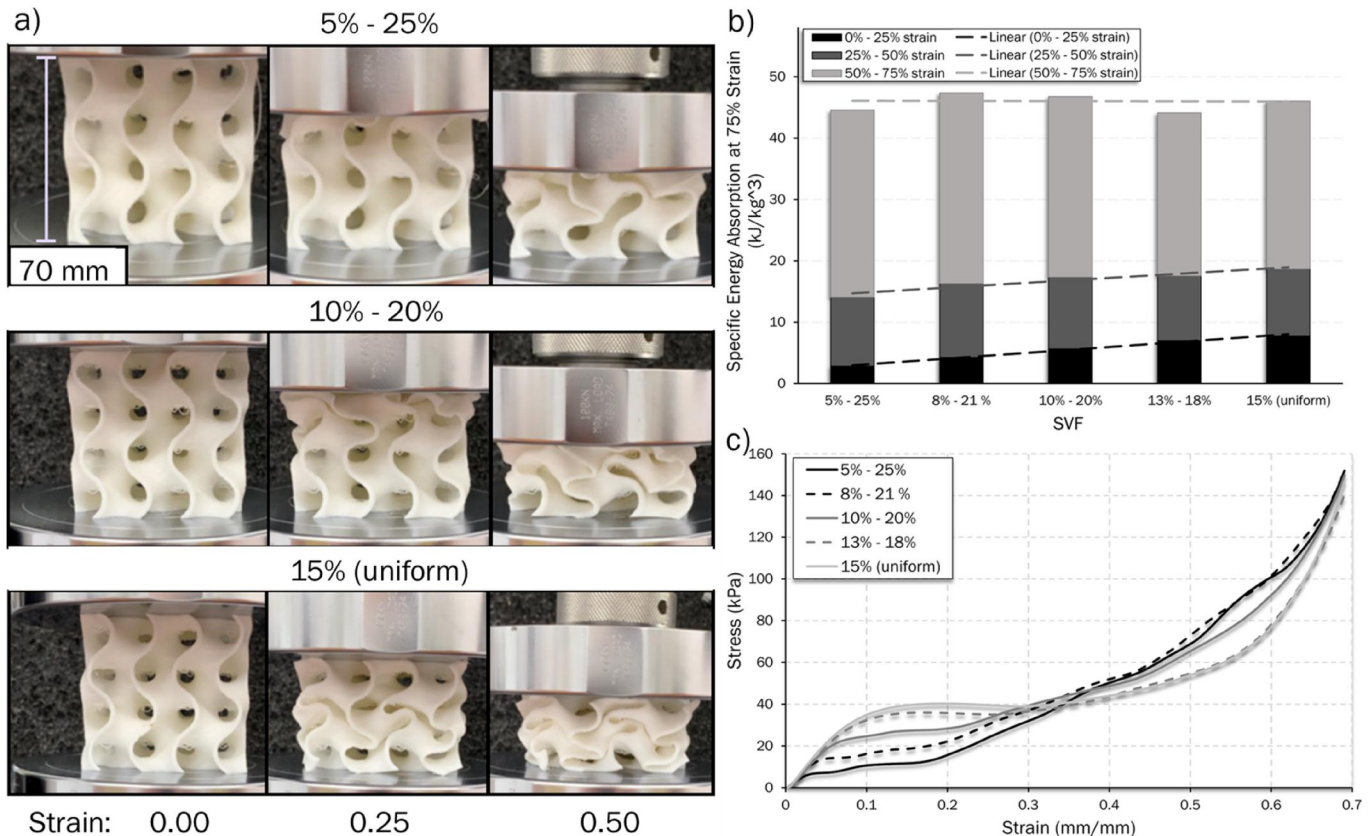


Figure 6. a) Comparison of the deformation of flexible gyroid structures with varying levels of density grading. Where applicable, the larger wall thickness is at the base of the structure, b) Impact of gradient steepness of specific energy absorption per unit volume and c) Compressive stress-strain response of functionally graded gyroids of differing stiffness gradients.

on a cushion with a higher vertical gradient, as would be desired.

3.3. Comparison to multilayered rehabilitation foam

The ability of two types of functionally graded gyroids, designed to replicate the mechanical properties of traditional multilayered rehabilitation foams, were evaluated by comparing their compression responses. This is compared in Figure 7a, where it can be seen for the multilayered foam, the soft Green EN36-90 foam at the top compresses almost to full densification before the Yellow EN40-230 and Pink MA35-600 foam regions begin to deform. In contrast, both gyroid samples deform more evenly across the total height in the direction of load. The stress–strain response in Figure 7b further shows that the gyroid design with distinct SVF regions has a slightly higher stiffness at low strains and lower stiffness at high strains as compared to the gyroid design with a linear SVF gradient. This shows the linear SVF gradient has a smoother curve than the sample with three separate stiffnesses, indicated by higher stress at low strains and lower stress at high strains, however there is not a significant difference in compressive behavior due to grading mechanism over the range observed here. The multilayered foam is far softer than both gyroid designs during the initial 5% of strain showing there are still manufacturing limitations with printing wall thicknesses low enough to match that of the softest rehabilitation foams. However, matching stiffness exactly during the first 5% of compression

is likely not critical to cushion performance. Additionally, Figure 7c shows the gyroid samples have lower stress than the multilayered foam at 10 and 50% strain with only the linearly graded gyroid having a higher stress at 70% strain. It follows that at both 50% and 70% strain, the multilayered foam exhibited the highest energy absorption with the gyroid sample with distinct SVF sections having the least. While the gyroid wall thickness was selected to replicate the different medical foams, it was observed that the achieved stiffness was slightly lower than that of the target foams. This is explained by the SVF of the manufactured samples being lower than targeted (Table 4 in Appendix A) and may be accounted for in future designs by adjusting target SVF to account for decreased print quality at low SVFs.

3.4. Scalability

A size effect analysis was performed to measure how the number of unit cells in a compression test sample impacted the results. Figure 8a shows the compressive stress–strain curve for three samples with varying number of unit cells in the direction of load. As few as one cell high was enough to capture a representative result for initial yield stress and plateau stress, however, densification occurred at a slightly lower strain. Beyond two cells high, no difference in the result could be observed. Therefore, sample height should capture a minimum of two unit cells, but one cell may be sufficient for some applications, including cushioning where the operating stress is typically within the plateau region.

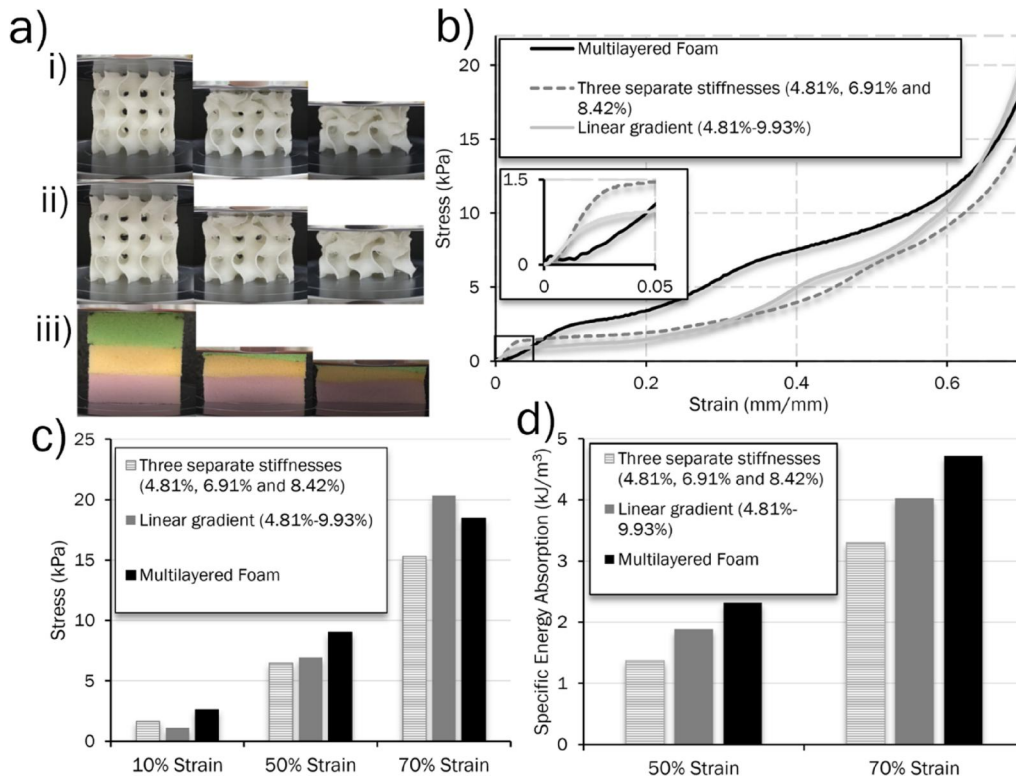


Figure 7. a) Compressive behavior of i) gyroid sample with three separate stiffnesses (4.81, 6.91, and 8.42% SVFs) and ii) gyroid sample with linear gradient (4.81–9.93% SVF) and iii) multilayered foam. b) Compressive stress–strain response, c) Comparison of stress values at different strain values and d) Comparison of specific energy absorption for functionally graded gyroids and multilayered rehabilitation foam.

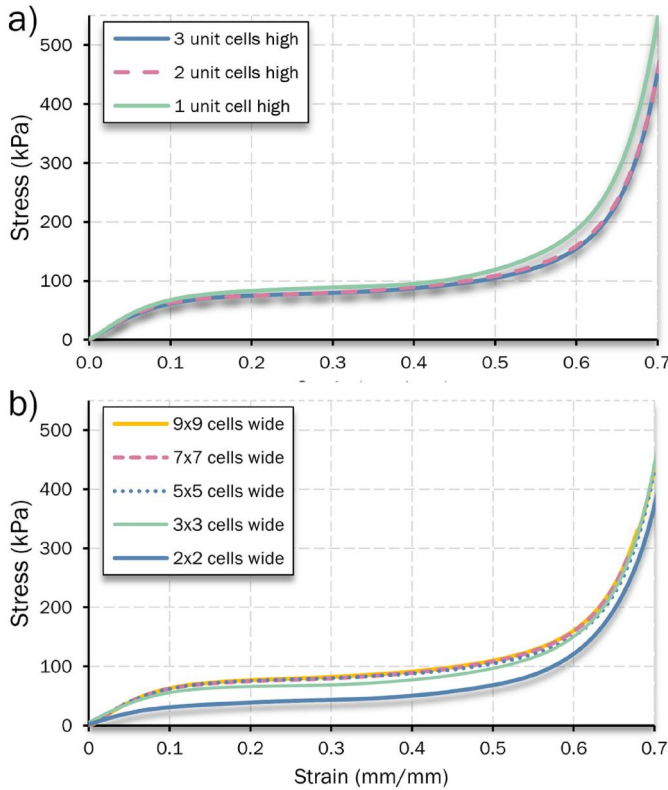


Figure 8. Compressive stress-strain response at different scales comparing the impact of a) number of unit cells in the direction of load (all 7×7 unit cells wide) and b) the impact of unit cells perpendicular to the direction of load (all 2 unit cells high).

The impact of sample width on compressive stress-strain behavior is shown in Figure 8b. Here 5×5 unit cells wide was found to be sufficient for obtaining scale independent lattice behavior. No statistically significant difference was observed between the plateau stress measured at 30% strain between the 9×9 , 7×7 and 5×5 unit cell samples. The plateau stress at 30% strain for the 3×3 cell sample was 16.4% lower than that of the 9×9 cell sample. It is unknown whether this difference is clinically significant. Therefore, it is possible that testing of samples with a width of 3 unit cells may be appropriate in some applications as long as the impact of scalability is considered.

3.5. Fatigue

Uniaxial compressive fatigue testing can give an indicator for durability of gyroid materials in the context of medical cushioning devices. The trend in stress-strain response was recorded for 10,000 cycles for structures with steep 5–25% SVF, and moderate 10–20%, gradients and a uniform 15% SVF control. Figure 9a shows the trend in peak stress across the first 10,000 cycles where the uniform sample appears to show the most softening and the sample with the largest gradient of 5–25% SVF had the least softening. This trend is thought to be largely due to the softening of the TPU base material from deformation-induced -heating. This effect is shown in the infrared images of each sample type taken immediately after the first 10000 cycles in Figure 9e. The initial temperature of the room and each sample ranged

from 22 to 24 °C. The uniform 15 and 5–25% SVF samples showed the most heating, up to a maximum of 38–39 °C, while the 10–20% SVF structure only reached a maximum temperature of 33 °C. The highest temperature regions appear to align with the regions of highest stress in the gyroid structure [45].

The samples were then rested for 24 h and retested for an additional five cycles until cycle 10,005. Table 3 shows the average peak stress for each sample for cycles 1, 5, 10,000, and 10,005. As it is standard to ignore the first four cycles when testing polymeric materials [43], the change between Cycle 5 and Cycle 10,005 is of the most interest here. This removes the influence of heating on the response and so best represents actual material degradation. The sample with the 5–25% SVF gradient had a decrease in peak stress of 14.2% from cycle 5 to 10,005. The 10–20% SVF structure lost only 8.5% while the uniform 15% SVF structure showed an increase in peak stress by 2.1%. These results seem to indicate the opposite trend to that across the first 10,000 loading cycles in Figure 9a. Figure 9b–d show the stress-strain curves at cycles 1, 5, 10,000, and 10,005. For the uniform 15% structure, the shape of the stress-strain curve changes significantly between cycle 5 and 10,005. The stress is lower at low strains and higher at high strains after fatigue loading, behaving closer to what is expected from FGMs. This indicates the top region of the structure typically often under the most strain is softening and effectively creating a vertical stiffness gradient in the structure. This effect is also seen for the graded structures to a lesser extent.

Considering all these factors, the 10–20% gradient structure appears to have been affected the least by fatigue loading of the designs observed here.

These findings emphasize that there are numerous factors impacting the fatigue resistance of 3D printed TPU structures. Looking at peak stress trends alone is a poor indicator of fatigue behavior. Utilizing a moderate vertical stiffness gradient appears to improve fatigue resistance in compression both in terms of peak stress and deformation behavior compared with uniform structures or those with large gradients. When considering the design of 3D printed cushions, vertical stiffness grading may improve cushion durability. However, for functionally graded 3D printed cushions, the uneven distribution of strain may cause some regions to fatigue faster than others, impacting the designed offloading effect. Because clinicians claim existing soft and moderate rehabilitation foams perform poorly in fatigue without layering of firmer foam underneath [46], future testing should measure the properties of traditional foams for a baseline to compare gyroids against.

3.6. Cyclic indentation testing

Cyclic indentation testing was performed up to 10,000 cycles for a foam structure and three designs previously shown in Figure 6, consisting of a uniform structure, a structure with a soft region directly under the indenter with instant transition to firmer material surrounding, and a structure where stiffness increased linearly across the structure radiating

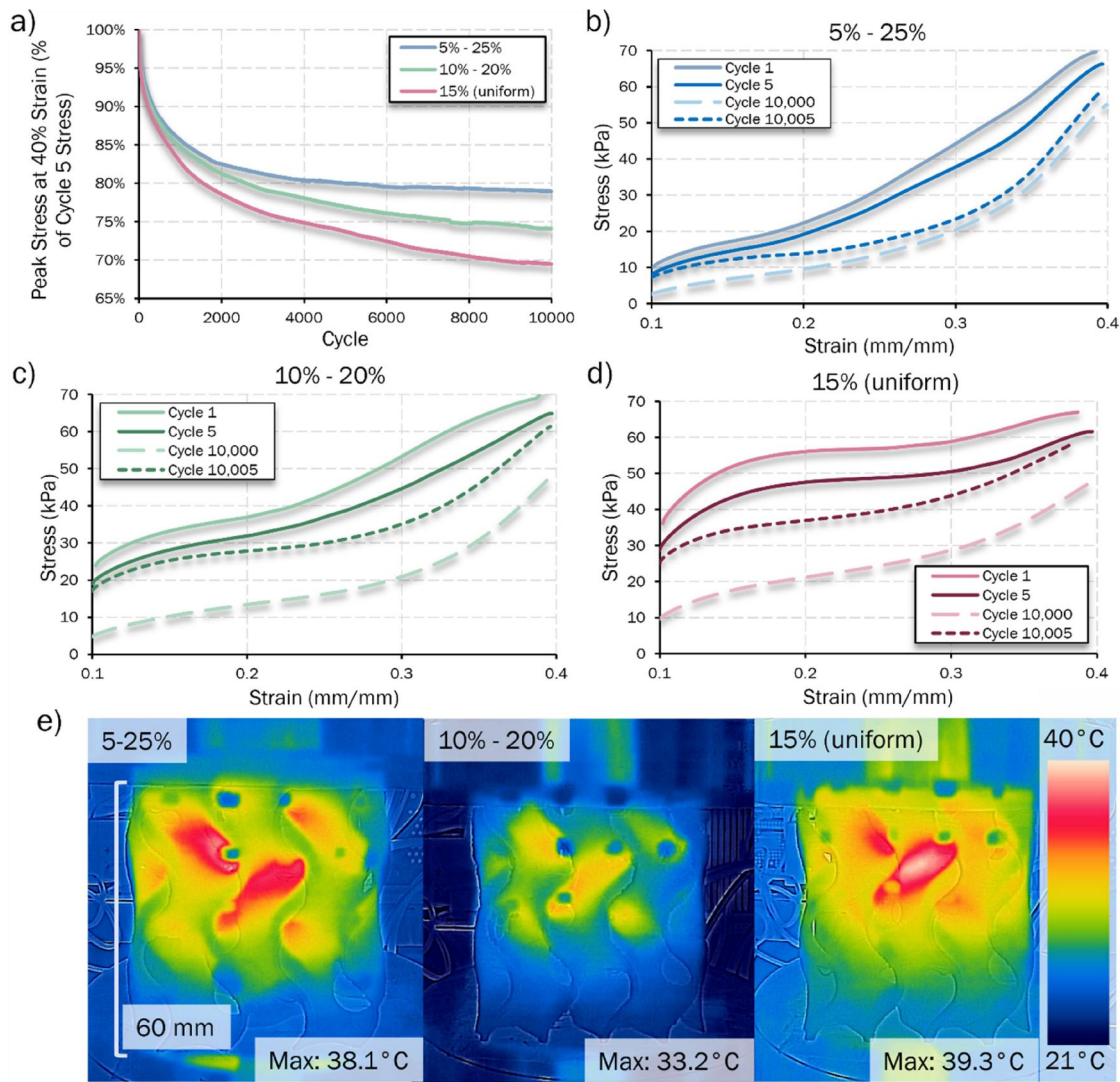


Figure 9. a) Trend in peak compressive stress over 10,000 cycles and trend in loading curve for b) 5–25% SVF graded gyroid structure, c) 10–20% SVF graded gyroid structure, d) 15% SVF uniform gyroid structure and e) infrared photos of the samples taken immediately after 10,000 cycles.

Table 3. Peak compressive stress at 40% strain at different cycles.

	Cycle 1 Stress (kPa)	Cycle 5 Stress (kPa)	Cycle 10,000 Stress (kPa)	Cycle 1,0005 Stress (kPa)	% Stress change from Cycle 5 to cycle 10,005
5–25%	70.7	66.8	55.0	59.0	–14.2%
10–20%	66.3	62.2	49.1	58.0	–8.5%
15% (uniform)	68.2	59.3	45.0	60.2	2.1%

Table 4. Peak stress at 40% strain at different cycles.

	Cycle 1 Force (N)	Cycle 5 Force (N)	Cycle 10,000 Force (N)	Cycle 10,005 Force (N)	% Force change from cycle 5 to cycle 10,005
Green EN36-90 Foam	30	28	24	23	–18.2%
6.91% Uniform	70	67	54	61	–9.5%
4.81–6.91% linearly increasing stiffness	55	53	43	49	–7.3%
4.81–6.91% instant transition between stiffness regions	52	50	42	45	–10.1%

from underneath the indenter. During testing, the room and initial sample temperature varied from 18 to 20 °C. In Figure 10, a thermal image taken at the end of 10,000 cycles shows the foam sample did not significantly change in temperature after loading, whereas the gyroid structures all increased slightly to 21–24 °C around the indenter.

Table 4 shows the peak stress at 40% strain for each sample after 1, 5, 10,000, and 10,005 cycles, where the sample

had been rested before the final cycle. All designs saw a significant drop in peak force from cycle 1 to cycle 5 and a further drop to cycle 10,000, which can also be seen in Figure 11. The trend in peak stress over the first 10,000 cycles shows the foam sample lost the least stiffness, while the uniform density gyroid structure lost the most. After resting and retesting at cycle 10,005, the gyroid samples regained some stiffness however the foam did not. From

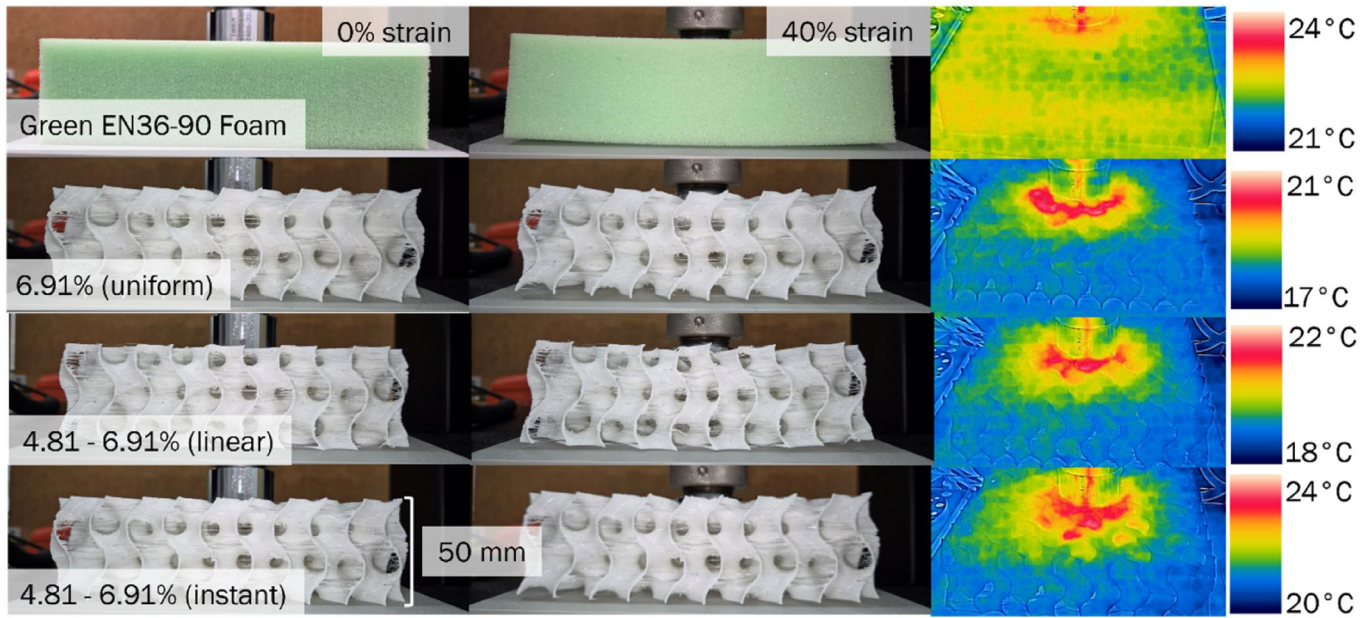


Figure 10. Visual compressive behavior of different samples during indentation testing (taken after 10,000 cycles).

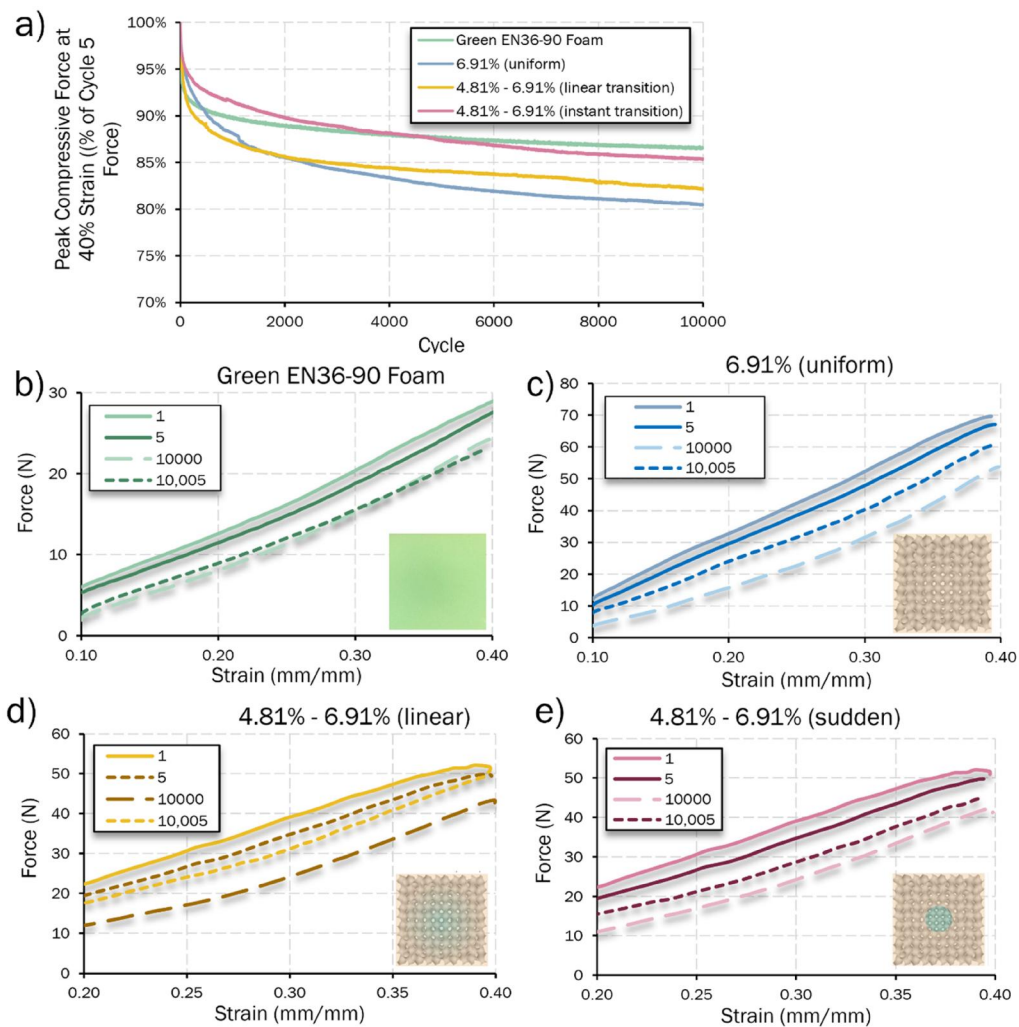


Figure 11. a) Trend in peak stress under 10,000 cycles of indentation to 40% strain, trend in loading curve for a) green EN36-90 foam, b) 6.91% SVF (uniform) gyroid structure, c) 4.81–6.91% linearly increasing SVF gyroid structure and d) 4.81–6.91% gyroid structure where SVF changes instantaneously.

cycle 5 to cycle 10,005, the loss in peak stress for the foam sample was 18.2%, and for the uniform 6.91% SVF gyroid structure, 9.5%. For the functionally graded gyroid structures, the design with linearly increasing SVF lost 7.3% while the design with the instant SVF change lost 10.1% peak stiffness. Fatigue resistance in the gyroid structures does not appear to have been negatively impacted by stiffness gradient here. The sample with the linearly increasing lateral gradient had the best fatigue resistance to indentation loading, followed by the sample with the instant stiffness transition. This was despite both these samples being softer than the uniform 6.91% SVF design and may indicate the structure more evenly distributed the load, reducing local damage. These findings suggest that lateral stiffness gradients in 3D printed cushion designs may improve fatigue resistance, in addition to providing an offloading effect. Smooth transitions, such as linear changes in wall thickness may also be favorable over instant changes.

4. Discussion

3D printed gyroid structures have been proposed as an alternative to traditional polyurethane medical foams in wheelchair cushions designed for reducing pressure injury risk. Cushions designed from such materials could have varying stiffness both vertically and laterally as required on a design-specific basis. The mechanical tests performed in this study were designed to improve understanding of how flexible gyroids may perform in this context.

First, Section 3.1 was designed to observe how grading the SVF across a gyroid structure impacted its compressive response. In Section 3.2, it was shown that a sample with SVF grading in the direction of loading saw lower stresses at low strains and higher stresses at high strains when compared with a structure without stiffness grading. The more SVF was changed, the greater this shift. This means under the same high stress, a cushion with a vertical stiffness gradient would deform less than a uniform cushion and be less likely to “bottom out.” These results verify the properties of graded gyroids behave as expected. Section 3.3 shows an example of design to replicate the properties of multilayered rehabilitation foams which further supported that flexible 3D printed structures have comparable mechanical response to compression as existing materials used in wheelchair cushions.

Section 3.4 tested the impact of the number of unit cells within a sample on the resultant compressive stress-strain response. It was found that samples with fewer unit cells tended to be softer than samples with more unit cells. This was because the edge effects are more impactful to the overall structure. The minimum number of cells required in the direction of load was two, although even one cell high was sufficient to represent response in the plateau stress region. A minimum of 5×5 unit cells wide was required to obtain a scalable result. Due to equipment limitations, it was not possible in this study to test samples of this size with 30 mm unit cells such as those which were in the range of comparable stiffness to rehabilitation foams. As such, it is expected that the results of Section 3.3 were impacted by edge effects,

and may be part of why the 3D printed foam design was softer than expected. This highlights the importance of considering the number of unit cells when mechanically testing metamaterials in future works.

Section 3.5 tested the impact of SVF gradient in the direction of loading on compressive fatigue and found fatigue resistance improved as the magnitude of the gradient increased. After 10,000 cycles, peak stress recorded decreased by 14.2% in a structure with a large SVF gradient and only 8.5% for a structure with a smaller SVF gradient. Peak stress for a uniform SVF structure increased after 10,000 cycles due to a damage induced change in the deformation behavior of the structure causing it to behave like an FGM. The results indicate that regions of FGMs under higher relative strains do soften more under cyclic loading, however this may cause stress at higher strains to increase meaning the whole stress-strain curve should be considered when evaluating the impact of fatigue. Self-heating also had a significant impact on the results highlighting the importance of allowing such materials to cool after high cyclic tests. Deformation induced heating of wheelchair cushions would not be expected in normal operating circumstances however heat induced softening resulting from the user's own body heat or operating in warm climates has not yet been considered and could be significant.

When SVF was varied in the direction perpendicular to loading and the sample was compressively indented for 10,000 cycles (Section 3.6), it was found that linearly increasing wall thickness provided better fatigue resistance compared with instantly transitioning wall thickness. The sample with linearly increasing lateral stiffness grading also outperformed a uniform stiffness structure, with peak stress recorded decreasing 7.3% compared with 10.1%. This means when designing functionally graded gyroid structures such as wheelchair cushions, smooth stiffness transitions may be favorable for cushion durability. The findings of both types of fatigue tests performed in this study found functional stiffness grading in TPU gyroid structures both vertically and laterally can improve overall fatigue resistance. However, if the minimum wall thickness in a design is too low, the part will fatigue earlier causing overall apparent stiffness to increase due to the bottoming-out effect. This highlights a need for high-cycle fatigue characterization of 3D printed foams before they can be used in cushions, as fatigue failure may potentially lead to adverse health outcomes.

The ability to 3D print structures with customized and variable stiffness within a structure introduces new flexibility in the design of wheelchair cushions which could not be achieved with traditional rehabilitation foams. The range of mechanical tests completed in this study support that flexible, functionally gyroid structures are suitable for use in wheelchair cushions as an alternative to foam.

4.1. Implications for clinical application

Previous studies on 3D printed TPU gyroids for cushioning applications suggest that these materials can be designed for a similar compressive response to existing rehabilitation foams [26–28]. Figure 12 shows a visualization comparing a multilayered foam cushion with a functionally graded 3D

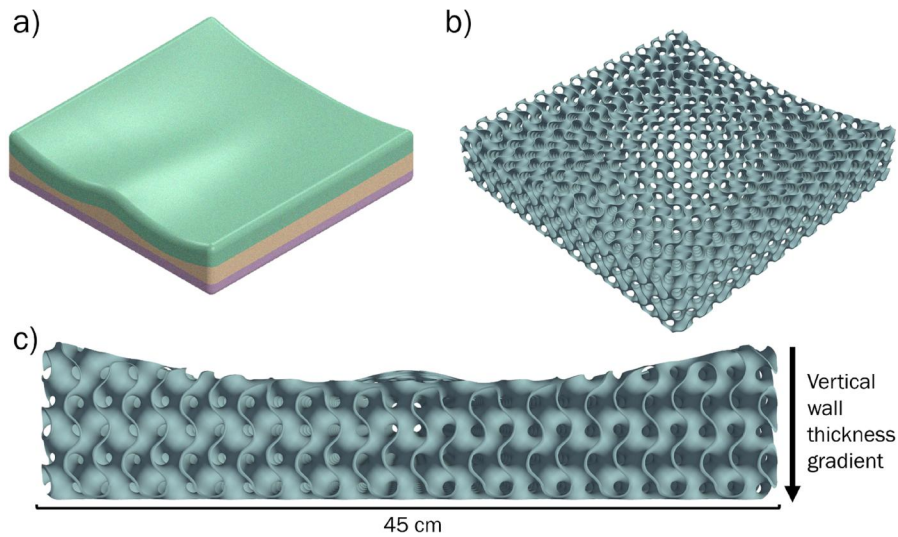


Figure 12. Visualizations of a) a contoured, multilayered foam cushion, b) a 3D printed cushion model designed to be mechanically equivalent to the foam cushion, c) front view of the 3D printed cushion showing the wall thickness gradient.

printed cushion. From top down, the foam cushion consists of three layers increasing in stiffness: Green EN36-90, Yellow EN40-230, and Pink MA35-600 foam manufactured by Dunlop Foams, Australia [1]. The 3D printed cushion also increases in stiffness the top to the base as SVF increases. Based on the results of this study, this design choice would improve the fatigue resistance and help prevent it from “bottoming out.”

Previous studies have recommended that the appropriate SVF range of TPU gyroid structures may fall between 4.8 and 15.5% for cushioning applications [26–28]. This finding was supported in this study, however, it has also been identified that small scale testing is not directly translational into application. A minimum number of 5×5 unit cells wide and 2 unit cells high must be tested to obtain scalable material properties. This is because results based on smaller samples than this will underestimate stiffness. Previous studies have only tested samples with a width covering 1–4 unit cells and so a cushion designed based on these results may be of a higher stiffness than desired. Therefore, neglecting to consider edge effects in design may negatively impact the comfort and effectiveness of the end product.

5. Conclusions and future work

This study investigated the effectiveness of functionally graded gyroids as an alternative for multilayered rehabilitation foams currently used as the gold standard of pressure injury prevention and treatment. It was found the mechanical properties of functionally graded, flexible gyroid structures are predictable and repeatable between studies and the compressive behavior of existing multilayered foams can be replicated and even improved upon using low-cost 3D printing methods. Through compression testing, it was shown functional stiffness grading both in the vertically and laterally can be used to extract desired mechanical response including cushioning and may translate to pressure offloading in wheelchair cushions and other medical devices.

Fatigue resistance was improved with vertical and lateral stiffness gradients and when using smooth transitions between different stiffness regions compared with sudden transitions. Future investigation should involve comprehensive fatigue testing of foams as compared with comparable gyroid structures as well as the impact of temperature on mechanical response. Most importantly, this work supports that flexible gyroid structures are a mechanically suitable alternative to rehabilitation foams to improve customizability in wheelchair cushion design. 3D printed wheelchair cushions have the potential to reduce waste, healthcare costs and pressure injury risk over traditional foam cushions. Future work should continue to build on the mechanical characterization performed in this work to inform and evaluate the design of 3D-printed cushions.

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Disclosure Statement

No potential conflict of interest was reported by the author(s).

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Appendix A

Printing results

The following Tables A1, A2, A3 and A4 show the variation between the target weight and what was measured for each sample after printing. Table A1 shows both functionally graded fatigue test samples printed just 1.6 and 1.67% lighter than designed while the uniform sample printed with 6.16% more material than designed. This indicates thinner walled samples are more susceptible to printing defects. As expected, the functionally graded samples used significantly more material than the one with constant wall thickness.

Table A1. Summary of sample design parameters and variations between designed and printed samples for gradient steepness study.

Sample	Initial thickness (mm)	Final thickness (mm)	Initial SVF	Final SVF	Equivalent overall target SVF $\frac{SVF_{initial} + SVF_{final}}{2}$	Actual mean SVF	Variation from target mean $\frac{actual - target}{target}$	Relative Standard Deviation of Actual SVF
5–25%	3	0.6	25.02%	5.05%	15.04%	14.40%	−5.05%	1.55%
8–21%	2.7	0.9	22.62%	7.66%	15.14%	14.42%	−5.41%	1.21%
10–20%	2.4	1.2	20.19%	10.21%	15.20%	14.64%	−4.19%	1.05%
13–18%	2.1	1.5	17.73%	12.73%	15.23%	14.41%	−5.84%	1.40%
15% (uniform)	1.8	1.8	15.24%	15.24%	15.24%	14.41%	−5.92%	0.49%

Table A2. Summary of sample design parameters and variations between designed and printed samples for gradient steepness study.

Sample	Average weight (g)	Equivalent overall target SVF $\frac{SVF_{initial} + SVF_{final}}{2}$	Actual mean SVF	Variation from target mean $\frac{actual - target}{target}$	Relative Standard Deviation of actual weight
Multilayered foam	8.18	—	—	—	3.24%
Gyroid with three separate stiffnesses, 4.81, 6.91, and 8.42%	17.01	6.62%	4.18%	4.78%	1.27%
Gyroid with linear stiffness gradient from 4.81 to 9.93%	16.64	6.72%	4.03%	−0.28%	0.43%

Table A3. Summary of sample design parameters and variations between designed and printed samples for scalability study.

Sample height (number of cells)	Sample width (number of cells)	Average weight (g)	Target SVF	Actual mean SVF	Variation from target mean $\frac{actual - target}{target}$	Relative Standard Deviation of actual weight
Impact of height						
1	7 × 7	10.29	17.84%	17.50%	−1.90%	0.35%
2	7 × 7	20.52	17.84%	17.45%	−2.21%	0.12%
3	7 × 7	30.95	17.84%	17.55%	−1.65%	0.18%
Impact of width						
2	2 × 2	1.63	17.84%	16.93%	−5.12%	0.31%
2	3 × 3	3.74	17.84%	17.31%	−2.94%	0.38%
2	5 × 5	10.46	17.84%	17.43%	−2.28%	0.08%
2	7 × 7	20.52	17.84%	17.45%	−2.21%	0.12%
2	9 × 9	34.04	17.84%	17.51%	−1.86%	0.06%

Table A4. Printing results of functionally graded gyroids for cyclic testing.

Sample	Target weight (g)	Printed weight (g)	Variation from target weight $\frac{actual - target}{target}$
6.91% SVF (uniform)	128.17	117.22	−8.54%
4.81–6.91% SVF (linearly increasing)	124.16	114.57	−7.73%
4.81–6.91% SVF (instant transition)	125.14	115.30	−7.86%